3 A fixed amount of an ideal gas of mass 0.35 kg has pressure 2.0×10^5 Pa and volume 0.038 m^3.

(a) In one process, the gas is heated at constant pressure so that its internal energy increases by 7600 J, volume increases by 0.025 m^3 and temperature increases by 56°C.

(i) Determine the thermal energy Q transferred to the gas.

$W_{on} = -p\Delta V = -(2.0 \times 10^5)(0.025)$
$= -5000$ J [C1]
$\Delta U = Q + W_{on}$
$7600 = Q + (-5000)$
$Q = 12600$ J [A1]

Q = ... J [2]

(ii) Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

$Q = mc\Delta T$
$12600 = 0.35c(56)$
$c = 640$ [A1] J kg^{-1} K^{-1} [A1]

specific heat capacity = ... unit ..... [2]

(b) In another process, the gas is heated at constant volume so that the increase in internal energy of the gas is the same as in **(a)**.

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in **(a)(ii)**.

No change in volume means no work is done, $W = 0$. [B1]

By first law of thermodynamics, $\Delta U = Q + W$, since ΔU remains the same and $W = 0$ instead of being negative, Q will be lower. The same gain in internal energy, ΔU implies the same increase in temperature. [M1]

Less thermal energy Q needed for same temperature change means a lower specific heat capacity. [A1]

.....[3]

[Total: 7]